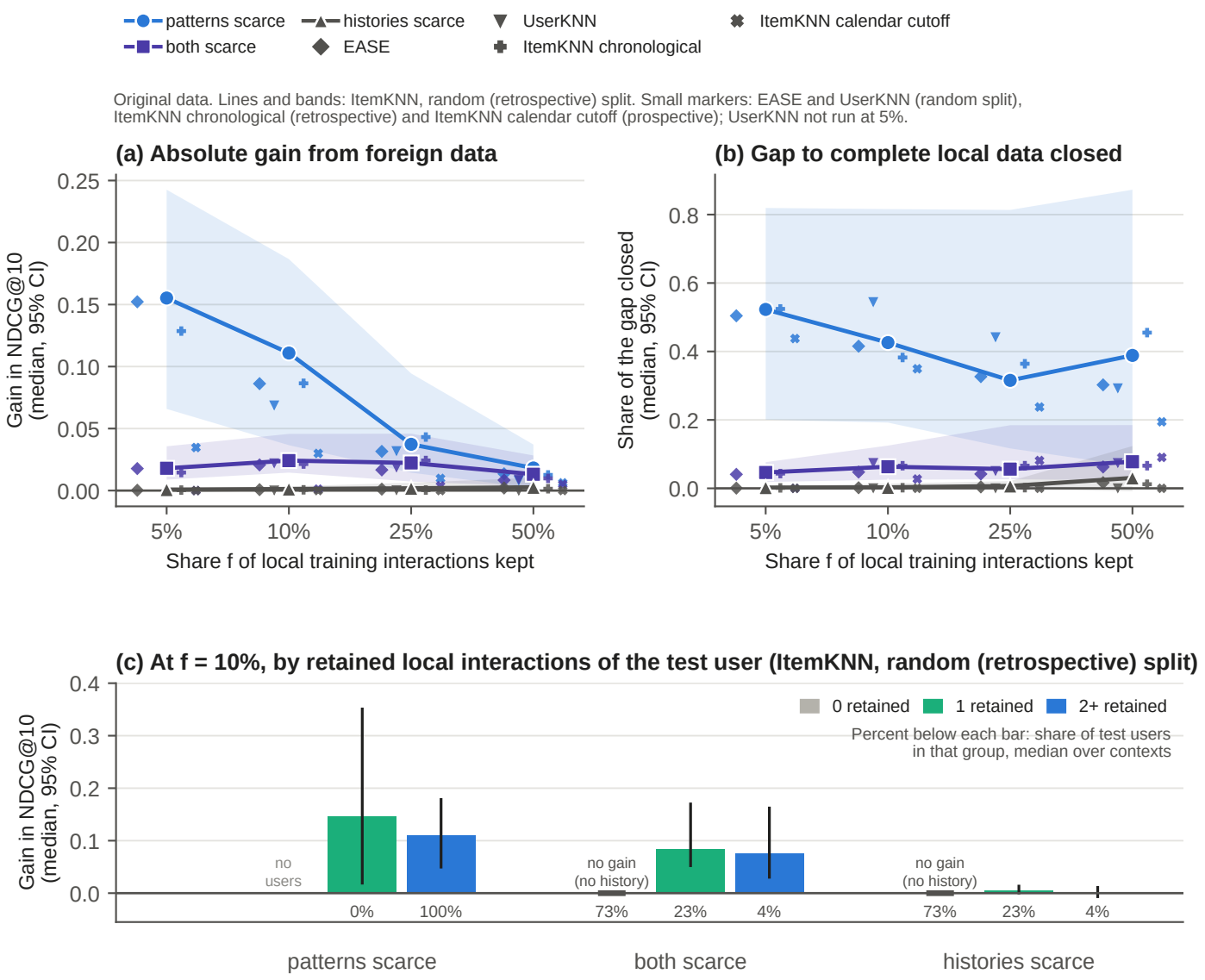
patterns scarce
histories scarce
UserKNN
ItemKNN calendar cutoff
both scarce
EASE
ItemKNN chronological
Original data. Lines and bands: ItemKNN, random (retrospective) split. Small markers: EASE and UserKNN (random split), ItemKNN chronological (retrospective) and ItemKNN calendar cutoff (prospective); UserKNN not run at 5%.
(a) Absolute gain from foreign data
Gain in NDCG@10 (median, 95% CI)
0.25
0.20
0.15
0.10
0.05
0.00
5%
10%
25%
50%
Share f of local training interactions kept
(b) Gap to complete local data closed
Share of the gap closed (median, 95% CI)
0.8
0.6
0.4
0.2
0.0
5%
10%
25%
50%
Share f of local training interactions kept
(c) At f = 10%, by retained local interactions of the test user (ItemKNN, random (retrospective) split)
Gain in NDCG@10 (median, 95% CI)
0.4
0.3
0.2
0.1
0.0
0 retained
1 retained
2+ retained
Percent below each bar: share of test users in that group, median over contexts
no users
0%
100%
no gain (no history)
73%
23%
4%
no gain (no history)
73%
23%
4%
patterns scarce
both scarce
histories scarce